

7 (a) (i) State what is meant by a photon.

.....
..... [1]

(ii) The first excitation energy of the hydrogen atom is 10.2 eV.

Explain what is meant by this statement.

.....
..... [1]

(b) Some electron energy levels of a particular atom are illustrated in Fig. 7.1.

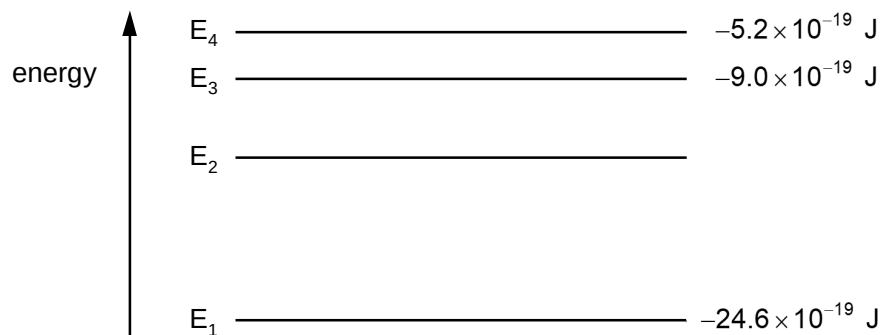


Fig. 7.1 (not to scale)

(i) Calculate the frequency of the electromagnetic radiation emitted when an electron makes a transition between energy levels E_3 and E_1 .

frequency = Hz [2]

- (ii) The frequency of radiation emitted when an electron makes a transition between energy levels E_3 and E_2 is 1.09×10^{15} Hz .

Determine the wavelength of the electromagnetic radiation when an electron makes a transition between energy levels E_2 and E_1 .

wavelength = nm [2]

- (c) An X-ray spectrum is shown in Fig. 7.2.

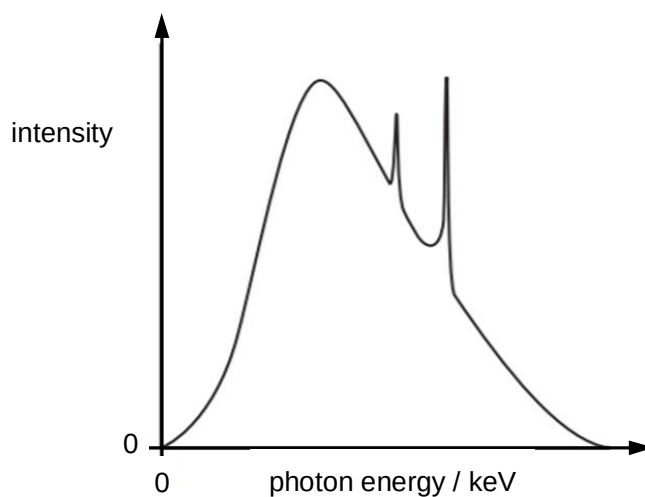


Fig. 7.2

Explain the process which gives rise to the characteristic peaks at certain photon energies.

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 [3]

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Section B

Answer **one** question from this section.

- 8 (a) State what is meant by simple harmonic motion.